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Slot: D

Course Name: C programming

1-

```c
#include <stdio.h>
int main()
{
    int distance, fare = 100, discount;
    int m;

    scanf("%d", &distance);
    scanf("%d", &m);

    if (distance > 10)
    {
        fare = fare + (distance - 10) * 20;
    }

    if (m == 1)
    {
        discount = fare * 0.10;
        fare = fare - discount;
    }

    printf("Final Fare: Rs.%d", fare);

    return 0;
}
```

2-

```c
**************************************************

#include <stdio.h>
int main()
{
    int q, t, s;

    scanf("%d", &q);
    scanf("%d", &t);
    scanf("%d", &s);

    if (q >= 1 && t > 0 && s == 0)
    {
        printf("Submission Allowed");
    }
    else
    {
        printf("Submission Not Allowed");
    }

    return 0;
}
```

3-

```c
#include <stdio.h>
int main()
{
    int treasure, x, y;
    int ben, blackbeard, remaining, pirate;
    scanf("%d", &treasure);
    scanf("%d", &x);
    scanf("%d", &y);
    ben = treasure * x / 100;
    remaining = treasure - ben;
    blackbeard = remaining * y / 100;
    remaining = remaining - blackbeard;
    pirate = remaining / 3;
    printf("%d\n", ben);
    printf("%d\n", blackbeard);
    printf("%d\n", pirate);
    return 0;
}
```

4-

```c
**********

#include <stdio.h>
int main()
{
    int d1, c1, d2, c2, total, dollars, cents;
    scanf("%d%d%d%d", &d1, &c1, &d2, &c2);
    total = (d1 * 100 + c1) + (d2 * 100 + c2);
    dollars = total / 100;
    cents = total % 100;
    printf("%d\n%d", dollars, cents);
    return 0;
}
```

5-

```c
**************************************************

#include <stdio.h>

int main()
{
    float mileage;
    int petrol, distance;
    scanf("%f%d%d", &mileage, &petrol, &distance);
    if (mileage * petrol >= distance)
    {
        printf("Can reach");
    }
    else
    {
        printf("Cannot reach");
    }
    return 0;
}
```

6-

```c
#include <stdio.h>
int main()
{
    int a, b, c, car;
    scanf("%d%d%d%d", &a, &b, &c, &car);
    if (a % car == 0)
    {
        printf("Car 1 goes into road A");
    }
    else if (b % car == 0)
    {
        printf("Car 1 goes into road B");
    }
    else if (c % car == 0)
    {
        printf("Car 1 goes into road C");
    }
    else
    {
        printf("No path exists");
    }
    return 0;
}
```

7-

```c
#include <stdio.h>

int main()
{
    int principal, rate, years;
    float interest, amount, discount, settlement;
    scanf("%d%d%d", &principal, &rate, &years);
    interest = principal * rate * years / 100;
    amount = principal + interest;
    discount = interest * 0.02;
    settlement = amount - discount;
    printf("%.2f\n", interest);
    printf("%.2f\n", amount);
    printf("%.2f\n", discount);
    printf("%.2f\n", settlement);
    return 0;
}
```

8-

```c
#include <stdio.h>
int main()
{
    int experience, performance;
    scanf("%d%d", &experience, &performance);
    if (experience > 7 && performance >= 85)
    {
        printf("Level: Senior");
    }
    else if (experience >= 3 && experience <= 7 &&
             performance >= 80)
    {
        printf("Level: Mid-Level");
    }
    else
    {
        printf("Level: Junior");
    }
    return 0;
}
```

9-

```c
#include <stdio.h>
int main()
{
    int year, month, days;
    scanf("%d%d", &year, &month);
    if (year < 1900 || year > 9999 || month < 1 || month > 12)
    {   printf("0");
    }
    else if (month == 1 || month == 3 || month == 5 ||
             month == 7 || month == 8 || month == 10 || month == 12)
    {   days = 31;
        printf("%d Days", days);
    }
    else if (month == 4 || month == 6 || month == 9 || month == 11)
    {   days = 30;
        printf("%d Days", days);
    }
    else if (month == 2)
    {
        if (year % 400 == 0 || (year % 4 == 0 && year % 100 != 0))
        {
            days = 29;
        }
        else
        {
            days = 28;
        }

        printf("%d Days", days);
    }
    return 0;
}
```

10-

```c
#include <stdio.h>
int main()
{
    int units;
    float bill;
    scanf("%d", &units);
    if (units <= 200)
    {
        bill = units * 0.5;
    }
    else if (units <= 400)
    {
        bill = units * 0.65 + 100;
    }
    else if (units <= 600)
    {
        bill = units * 0.80 + 200;
    }
    else
    {
        bill = units * 1.25 + 425;
    }
    printf("Rs.%.0f", bill);
    return 0;
}
```